

DUTCH HARBOR, AK, USA

WMO: 704890

Lat: 53.895N

Lon: 166.543W

Elev: 10

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25616

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	21.9	10.0	9.2	21.6	12.3	10.3	23.5	38.1	37.3	34.5	34.4	15.1	330	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	63.4	57.7	60.7	55.8	57.4	52.9	58.4	62.8	56.0	60.1	53.7	57.0	11.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.3	65.2	61.6	53.7	61.3	59.4	51.7	57.0	56.9	25.2	62.7	23.7	60.1	22.4	57.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.7	28.9	25.8	17.3	70.7	5.3	3.8	13.5	73.4	10.3	75.6	7.3	77.8	3.5	80.5
			15.5	63.7	4.7	3.0	12.1	65.8	9.4	67.6	6.8	69.2	3.4	71.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	33.7	34.5	34.4	37.0	41.4	46.8	51.4	53.4	49.5	43.5	38.7	35.4	(6)
(7)		DBStd	8.03	5.29	5.16	5.13	4.06	3.42	2.80	3.09	3.75	3.56	4.05	4.23	4.97	(7)
(8)		HDD50	3253	504	433	485	390	266	104	18	7	49	204	339	454	(8)
(9)		HDD65	8514	969	853	950	840	731	547	423	362	465	666	789	919	(9)
(10)		CDD50	216	0	0	0	0	0	7	60	111	35	3	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0	(11)
(12)		CDH74	1	0	0	0	0	0	0	0	1	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	11.1	12.0	12.3	11.9	12.2	10.2	8.7	8.0	8.5	11.2	12.6	13.2	12.7	(14)
(15)	Precipitation	PrecAvg	69.3	8.2	6.8	5.8	3.7	4.4	2.6	2.6	3.0	5.9	7.6	7.3	8.9	(15)
(16)		PrecMax	86.4	16.9	14.5	14.7	6.7	10.2	4.1	7.3	6.2	10.0	18.0	19.6	18.9	(16)
(17)		PrecMin	49.6	2.2	1.2	1.5	1.5	1.1	0.7	0.2	0.5	1.1	4.3	2.8	1.9	(17)
(18)		PrecStd	10.7	3.5	3.1	3.7	1.5	2.1	1.1	1.6	1.5	2.7	3.1	4.3	4.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	46.8	48.1	48.2	52.5	59.5	66.5	69.9	63.0	55.4	51.9	48.4	(19)
(20)			MCWB	40.8	41.9	44.1	44.2	46.8	53.3	60.5	63.1	58.1	50.8	46.9	44.1	(20)
(21)		2%	DB	43.2	44.3	44.3	45.4	48.7	54.7	62.1	64.4	59.1	52.8	48.4	45.9	(21)
(22)			MCWB	39.6	40.1	40.1	40.9	43.7	49.6	56.5	58.9	54.7	48.2	44.7	42.3	(22)
(23)		5%	DB	41.4	42.5	42.4	43.9	47.3	52.4	58.2	61.7	56.5	50.8	46.3	43.7	(23)
(24)			MCWB	38.4	38.8	38.5	39.4	43.0	47.9	53.6	56.6	52.1	46.5	42.7	40.0	(24)
(25)		10%	DB	39.8	40.5	40.5	42.7	45.8	50.8	55.6	58.8	54.6	48.9	45.0	41.8	(25)
(26)			MCWB	36.8	37.3	37.1	38.6	41.9	46.7	51.6	54.5	50.2	45.0	41.1	38.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	42.4	44.3	44.5	47.1	54.3	61.1	63.4	58.6	51.6	47.3	45.2	(27)
(28)			MDCB	44.7	45.8	47.9	47.7	51.4	58.8	66.2	68.7	62.4	55.0	50.7	48.0	(28)
(29)		2%	WB	40.4	40.8	40.7	41.5	45.1	50.3	56.8	59.5	55.0	48.7	45.6	42.7	(29)
(30)			MDCB	43.1	43.7	43.8	44.9	47.7	54.0	61.3	64.0	58.6	52.1	48.5	45.6	(30)
(31)		5%	WB	39.0	39.2	39.0	40.2	43.6	48.3	54.2	57.2	52.6	47.0	43.4	40.7	(31)
(32)			MDCB	41.7	41.9	42.0	43.5	46.6	51.6	57.6	61.5	56.0	50.0	46.2	43.7	(32)
(33)		10%	WB	37.6	37.7	37.4	39.0	42.3	47.0	52.1	55.0	50.8	45.7	41.5	39.0	(33)
(34)			MDCB	39.7	40.1	40.2	42.2	45.5	49.9	55.2	58.6	54.1	48.5	44.8	41.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.3	6.2	5.9	6.1	6.2	7.4	7.5	6.2	5.8	6.2	5.9	(35)
(36)			MCDBR	7.1	8.9	8.1	7.7	8.4	9.5	12.3	12.3	9.2	7.4	8.2	7.9	(36)
(37)			MCWBR	6.2	7.0	6.8	6.1	5.9	6.2	8.7	8.9	7.2	6.3	7.6	6.9	(37)
(38)		5% WB	MCDBR	7.0	7.3	7.7	7.3	7.4	8.7	12.0	11.9	8.8	6.8	8.0	7.8	(38)
(39)			MCWBR	6.3	6.7	6.9	6.1	5.6	6.1	9.1	9.1	7.3	6.3	7.8	6.9	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.310	0.346	0.354	0.402	0.400	0.393	0.392	0.393	0.375	0.353	0.334	0.305	(41)	
(42)		taud	2.244	2.215	2.256	2.148	2.238	2.346	2.407	2.399	2.360	2.294	2.185	2.202	(42)	
(43)		Ebn at Noon	201	231	259	260	268	272	270	262	251	226	184	174	(43)	
(44)		Edh at Noon	20	28	33	42	41	37	34	33	30	26	21	17	(44)	
(45)	All-Sky Solar Radiation	RadAvg	173	369	738	1105	1332	1398	1222	1058	814	505	227	130	(45)	
		RadStd	19	31	73	84	115	104	132	80	56	35	22	14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page